ME 617 Mock Lab Assignment

Scanned-Wavelength Direct-Absorption Measurements of CO, Temperature, and Pressure in a High-Temperature Shock Tube experiment

Introduction: The primary goal of this lab is to illustrate how laser-absorption spectroscopy can be used to characterize a gas with an unknown temperature and chemical composition. You will use scanned-wavelength direct-absorption spectroscopy to measure the gas temperature and mole fraction of carbon monoxide in a shock-heated mixture of CO-CO_2-N_2. In doing so, you will see many of the key concepts of ME 617 in action. Further, interpreting the experimental data we acquire will require you to utilize many of the theoretical concepts we have discussed in class.

Experimental Procedure and Equipment

High-Pressure Shock Tube (HPST): Laser-absorption measurements of high-temperature CO will be acquired in Purdue's High-Pressure Shock Tube (HPST). A shock tube is a device used to near instantaneously generate high-temperature and high-pressure gas conditions to enable the study of a wide range of physical phenomena (e.g., combustion, molecular energy transfer, spectroscopy, etc.). A shock tube consists of a long, enclosed tube which is divided into a driver and driven section by a diaphragm (see Fig. 1). The driven section is filled with the test gas to be studied, in this case, a mixture of 2% CO, 1.8% CO_2, with a balance of N_2. The driver section is pressurized until the diaphragm bursts, creating a shock wave which propagates down the driven section and reflects off the endwall, creating a region of stagnant, high-temperature, higher-pressure gas, which is typically the region of interest. Gas conditions behind the incident and reflected shock waves are denoted as "Region 2" and "Region 5", respectively.

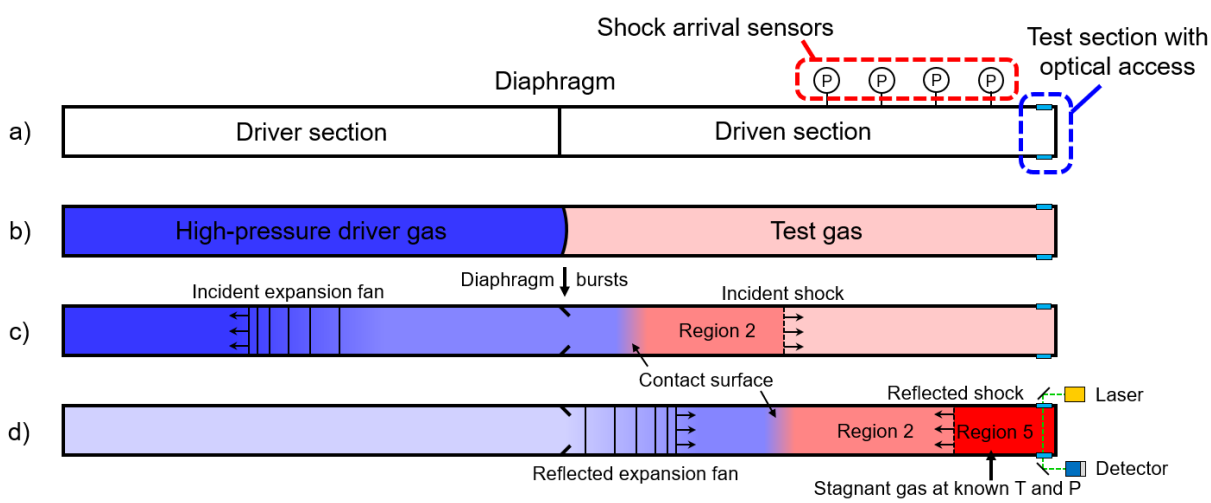


Figure 1: A schematic showing the stages of a typical shock-tube experiment.

Laser absorption measurements were acquired at the far end of the driven section (often called the test section) which contains windows to allow access for various optical diagnostics. A high-speed pressure transducer (Kistler) measures pressure in the test section. Shock arrival sensors are used to measure the velocity of the incident shock wave, and enable determination of gas conditions in Region 2 and Region 5 using shock-jump equations. Region 5 ends when the expansion fan from the driver end of the shock tube arrives at the measurement location (determined by the Kistler) or once an unacceptable rise in pressure (e.g., from a reflection off the contact surface or boundary layer growth) is observed at the measurement location. This experiment targeted a Region 5 temperature (T_5) of 4000 K and 1 atm, respectively. The internal diameter of the shock tube is 10.32 cm.

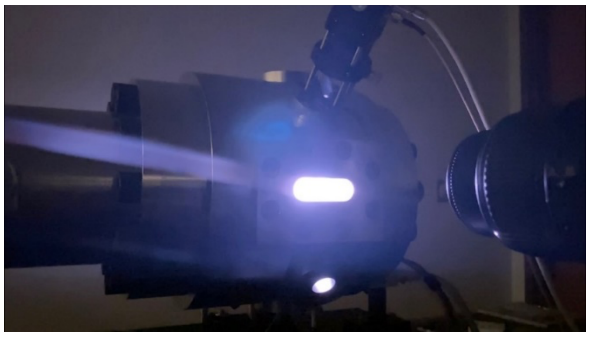


Figure 2: Left: Photo of gas in Region 5 during a shock-tube experiment with gas temperatures near 5000 K! Right: Mugatu determines that T_5 is high from his fundamental understanding of optical emission processes.

Laser-Absorption Spectroscopy Diagnostic:

Figure 3 illustrates the primary components of your experimental setup (according to a sarcastic postdoc). A quantum-cascade laser (QCL) with a wavelength near 4.98 μm was used to measure the absorbance spectrum of the P(0,31), P(2,20), and P(3,14) absorption transitions of CO where the transition nomenclature corresponds to P(v", J"). The QCL (Alpes Lasers) produces a collimated laser beam (≈1.5 mm diameter) with an optical power of 100 mW and an instantaneous linewidth of approximately 5 MHz. The nominal temperature and current of the laser are controlled using a commercially available laser controller (Wavelength Electronics). A unique attribute of many semiconductor lasers (e.g., diode lasers,

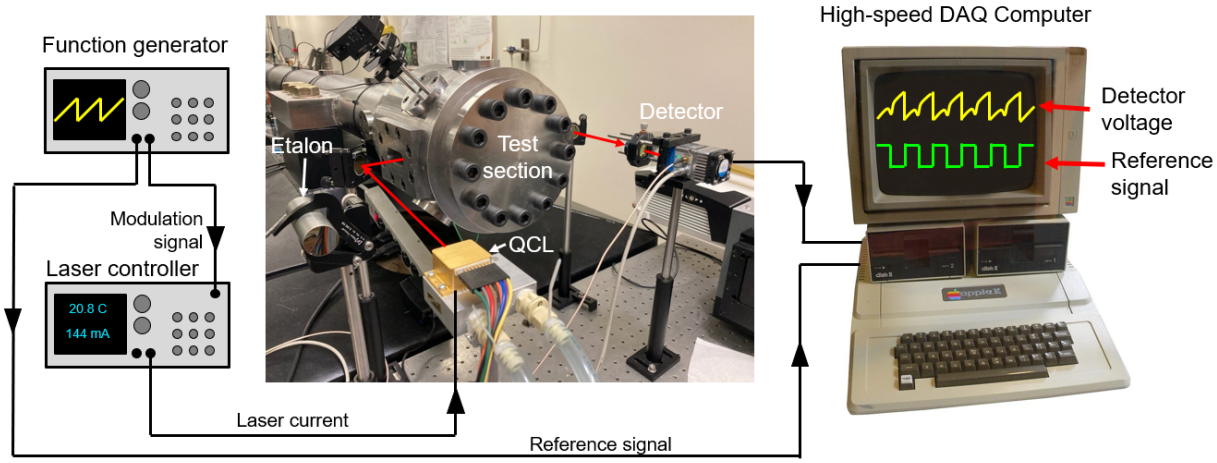


Figure 3: Schematic and photograph of the experimental setup used to acquire scanned-DA measurements of temperature, pressure, and CO in a shock tube experiment.

interband-cascade lasers, quantum-cascade lasers) is that their wavelength and intensity (i.e., power) can be tuned near-linearly by modulating the laser's current. In our experiments, the QCL's current was modulated in time by supplying a voltage modulation (sawtooth wave, 0 to 620 mV, 300 kHz) to the laser controller's modulation port. The triangle wave was generated using an analog function generator (Tektronix).

The laser beam was directed through a pair of windows in the HPST's test section and focused onto a photovoltaic, AC-coupled, MCT (mercury cadmium telluride) detector (Vigo Systems) using an anti-reflection coated, plano-convex, CaF_2 lens (Thorlabs, 25.4 mm diameter, 40 mm focal length). The photodetector is sensitive to wavelengths from approximately 3.5 to 11 μm and produces a voltage output

that exhibits a linear dependence on the optical power collected. The electrical bandwidth of the detector is 200 MHz. The photodetector's voltage signal was recorded using a 12-bit data-acquisition (DAQ) system (GaGe) with a bandwidth of 500 MHz and a sampling rate up to 3 GS/s. Onboard averaging of the detector signal was performed to record the detector signal at a sampling rate of 375 MHz and an effective bit depth of 14 bits for improved signal-to-noise ratio.

An additional experiment was conducted to characterize how the wavelength of the laser changes in time in response to the current modulation. A solid germanium etalon was placed in the beam path. The etalon is a 3" long rod with polished ends, designed such that the primary laser beam (passing through the etalon) is collinear with a reflected beam (formed at the air-germanium interface) that is passing backwards through the etalon. The superposition of coherent laser beams (i.e., electromagnetic waves) that are phase-shifted relative to one another leads to an interference pattern in the detected laser beam. As the wavelength of the laser is change in time, the amount of constructive and destructive interference in the transmitted laser beam varies in time. Resonances occur whenever an integer number of wavelengths fit within the length of the etalon. The spacing between resonant peaks is the free-spectral range (FSR) of the etalon, given by:

$$\mathrm{FSR} = \frac{1}{2\mathrm{nL}}$$

Here, L is the length of the etalon in (cm) and n is the refractive index of germanium at the laser wavelength (n = 4.019 at 4.98 μm).

Deliverables: You will be provided with experimental data (detector signal vs time) for 3 experiments: 1) data from the shock tube experiment, 2) baseline data with no CO present, and 3) etalon data. Each data file contains the detector signal and a TTL (reference signal) signal from the function generator. You will use the reference signal to precisely align different datasets.

You must determine how the temperature, pressure, and CO mole fraction vary in time during the shock tube experiment. Each student must submit: 1) a copy of the code(s) (e.g., m file) they personally developed and used to process the experimental data and 2) a 5-7 page technical report describing: i) the relevant theory, ii) the experimental setup and procedure, iii) the experimental results and findings, iv) the measurement uncertainty, and v) conclusions. Your report should be written single-spaced with 1" margins and 12-point Times New Roman font. Your report will be graded based on technical accuracy and quality and clarity of presentation.

Spectroscopic Parameters: Below are the pertinent spectroscopic parameters for the absorption transitions of interest.

Table 1: Relevant spectroscopic parameters taken from the HITEMP 2019 database for CO and Hartmann et al. 1988.

Transition	ν_0 [cm^{-1}]	S_{ref} (T_0 = 296 K) [cm^{-1}/(molec-cm^{-2})]	**E"** [cm^{-1}]	γ_{CO-N_2} [cm^{-1} /atm]	n_{N_2} [-]	γ_{CO-CO} [cm^{-1} /atm]	n_{CO} [-]
P(0,31)	2008.525	2.669e-22	1901.131	0.0412	0.47	0.0430	0.5
P(2,20)	2008.422	1.149e-28	5051.740	0.0526	0.57	0.0550	0.5
P(3,14)	2008.552	2.877e-32	6742.874	0.0607	0.65	0.0610	0.5

Tutorials: The figures below illustrate the primary data-processing steps you must take to determine a single measurement of temperature and CO mole fraction. In addition, I have provided you with several m-files that demonstrate how to determine the best-fit spectrum and corresponding spectroscopic parameters for a similar absorption spectroscopy measurement. You must incorporate these steps into your own code that can process a time history of data spanning Regions 2 and 5 of the shock tube experiment (≈ 150 scans).

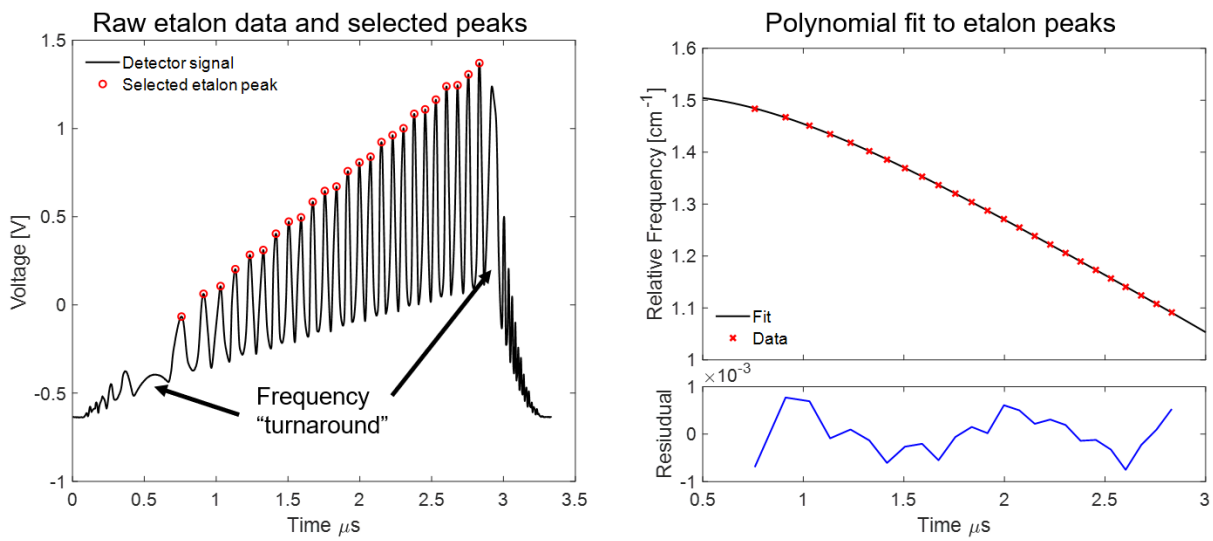
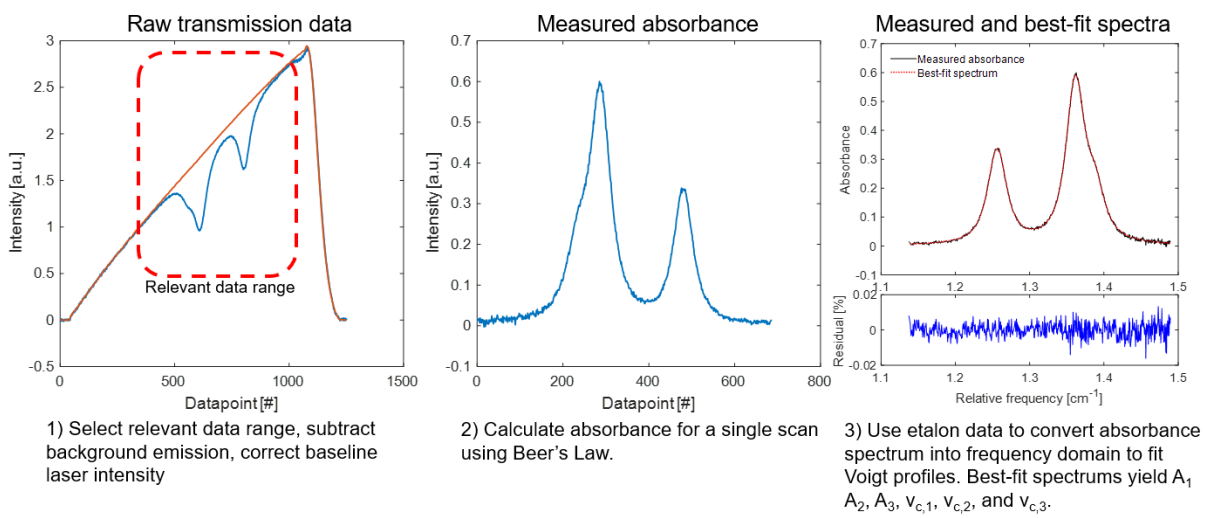


Figure 2: Key steps involved in processing etalon data.



1) Select relevant data range, subtract background emission, correct baseline laser intensity

2) Calculate absorbance for a single scan using Beer's Law.

3) Use etalon data to convert absorbance spectrum into frequency domain to fit Voigt profiles. Best-fit spectrums yield A_1 A_2, A_3, $v_{c,1}$, $v_{c,2}$, and $v_{c,3}$.

Figure 3: Key steps involved in processing shock tube experiment transmission data

HINTS:

1) In your spectral-fitting routine, you can assume the Doppler width is constant and equal for both transitions. As a start you can assign it a value corresponding to 3750 K and then refine it after you acquire your first successful temperature time history.
2) Remember the key concept behind processing absorption spectroscopy data, which is to always have fun.
3) The broadening half-widths provided were determined from fitting a model to at collected at up to 1850 K. This means that trusting the accuracy of these parameters at 4000 K is a dubious

proposition. I recommend scaling the collisional widths by 0.84 when using them to calculate pressure.

4) The comparatively weak P(3,14) transition may lead to difficulty in getting accurate fit for this transition, which may introduce errors into the parameters form the other two lines. To make the fitting routine more robust you may constrain $\nu_{c,\,P(3,14)}$ to be equal to $\nu_{c,\,P(2,20)}$ in the fitting routine. Likewise, you may find it helpful to set $\nu_{0,\,P(3,14)}$ as a fixed offset from $\nu_{0,\,P(0,31)}$.
5) If you have two computer monitors, always keep spreadsheets or MATLAB open on one monitor to give the illusion that you are hard at work.
6) You can assume the same broadening coefficients for CO-CO_2 broadening as for CO-N_2 broadening.

Good luck!